

To create a ready training dataset in Excel that matches tree data to plots and scales plot-level sums to per hectare values, the National Forest Inventory (NFI) report provides detailed methodologies and data structures, which serve as the basis for a structured workflow as follows:

Step 1: Import Data into Excel

- Create two worksheets named *Plots* and *Trees*.
- In *Plots*, paste plot data with key columns including EI BlockID, Transect no, Plot No, Date, plot_coordinate_x, plot_coordinate_y, and altitude.
- In *Trees*, paste tree-level data with EI BlockID, Transect no, Plot No, Tree no, stump, point_of_measurement, stump_age, Species code, Species name, DBH (Diameter at Breast Height in cm), Height (m), among others.

This setup reflects the data entities described in section 3.5 and Tables in the report, which provide the foundational data structure for the NFI.

Step 2: Prepare Data for Matching

- Ensure EI BlockID, Transect no, and Plot No columns have consistent formatting (e.g., no trailing spaces) to facilitate accurate matching between *Plots* and *Trees*.
- Sort both worksheets by EI BlockID, Transect no, Plot No, and Tree no.

Given the spatial layout of the sampling design—plots nested within blocks and transects—uniform data formatting is essential for correct linkage.

Step 3: Match Trees to Plots

- Generate a unique PlotID in both worksheets by concatenating EI BlockID, Transect no, and Plot No, for example:

```
=A2 & "-" & B2 & "-" & C2
```

This mimics the geo-referencing of plots as identified by block and transect coordinates in the inventory (sections 3.1.1 and 3.4.2).

- Use PlotID as the key to associate individual trees with their respective plots.

Step 4: Calculate Plot-Level Tree Metrics

- Create a *PlotSummary* worksheet listing all unique PlotIDs.
- Use lookup formulas (e.g., `VLOOKUP` or `INDEX-MATCH`) to retrieve plot-level attributes from *Plots*, such as coordinates and altitude.
- Compute plot-level metrics using Excel aggregation formulas like:
- Tree Count per plot:

```
=COUNTIF(Trees!$E:$E, A2)
```

- Sum of DBH (cm):

=SUMIF(Trees!\$E:\$E, A2, Trees!\$K:\$K)

- Average Height (m):

=AVERAGEIF(Trees!\$E:\$E, A2, Trees!\$L:\$L)

Here, column references assume PlotID is in column E and DBH in column K per *Trees* sheet.

This follows the approach to aggregating tree attributes to plot level prior to scaling, consistent with sections 4.2 and 6 of the report.

Step 5: Compute Basal Area and Biomass Per Tree (Optional Detailed Step)

- Compute basal area per tree to represent cross-sectional area at breast height (m²) using:

Basal Area= $\pi \times (200\text{DBH})^2$

- Implement this in Excel as:

=PI() * (K2 / 200)^2

where K2 is DBH in cm.

- Sum basal area at plot level:

=SUMIF(Trees!\$E:\$E, A2, Trees!\$<BasalAreaColumn>)

- Estimate above-ground biomass (AGB) per tree using allometric equations that require DBH, height, and wood density (WD), per Chave et al. (2014):

AGB= $\exp(a+b \times \ln(\text{DBH})+c \times \ln(\text{Height}))$

- For EI data with missing height measurements, heights are estimated from DBH using site-specific height-diameter models (Curtis 1967), as described in section 4.1.1.2 with parameters provided in Table 4-2.
- Compute below-ground biomass (BGB) as a fraction of AGB using root to shoot ratios per land strata from IPCC guidelines (Table 4-3).
- Total biomass (TB) is the sum of AGB and BGB:

TB=AGB+BGB

- Tree carbon stock is derived by multiplying biomass by a conversion factor (typically 0.5).

Step 6: Scale Plot-Level Sums to Per Hectare Basis

- Recognize that plot sizes vary by inventory type: EI plots are 0.05 ha, NBS plots 0.25 ha, with nested plots smaller (EI nested plot = 0.0125 ha).
- Apply expansion factors to convert raw plot measurements to per hectare estimates:
- For EI plots: multiply sums by 20 (1 ha / 0.05 ha).
- For EI nested plots: multiply by 80.
- For NBS plots: multiply by 4.
- For NBS nested plots: multiply by 20.
- For example, per hectare basal area is:

= BasalAreaSum * 20 (for EI plots)

- Similarly for tree counts, biomass, carbon stocks.

This approach aligns with the harmonization and data transformation techniques detailed in section 4.1.1.1.

Step 7: Finalize and Export Training Dataset

- Compile columns in *PlotSummary* including:
- PlotID, EI BlockID, Transect no, Plot No
- Date, plot_coordinate_x, plot_coordinate_y, altitude
- Tree Count per hectare
- Total DBH per hectare
- Basal Area per hectare
- Average Height
- Biomass and Carbon stock estimates (if calculated)
- Check for missing values and ensure every PlotID with trees has corresponding plot information.
- Save the Excel file in .xlsx format for use in training or modeling workflows.

Summary Table of Key Excel Formulas and Data Columns

Action	Excel Formula Example	Notes
Create Unique PlotID	=A2 & "-" & B2 & "-" & C2	Concatenate EI BlockID, Transect no, Plot No
Lookup plot coordinate X	=VLOOKUP(A2, Plots!\$G:\$J, 2, FALSE)	A2 = PlotID, returns plot_coordinate_x
Tree count per plot	=COUNTIF(Trees!\$E:\$E, A2)	Count number of trees with PlotID = A2
Sum DBH per plot	=SUMIF(Trees!\$E:\$E, A2, Trees!\$K:\$K)	DBH in column K
Calculate Basal Area/tree	=PI() * (K2 / 200)^2	DBH in cm, basal area in m ²
Sum Basal Area per plot	=SUMIF(Trees!\$E:\$E, A2, Trees!\$<BasalAreaCol>)	Aggregate basal areas per plot

Action	Excel Formula Example	Notes
Scale to per hectare (EI)	Multiply sums by 20	EI plots are 0.05 ha; expansion factor = 20

Rationale and References from the Report

- **Data Structure & Sampling Design:** The use of EI and NBS inventories with different plot sizes and shapes necessitates harmonization of data and use of expansion factors.
- **Plot Demarcation & Measurement Protocols:** Detailed procedures for plot location, demarcation, and measurement ensure standardization vital for accurate data matching.
- **Computation of Tree Metrics:** Basal area, biomass, and carbon calculations are essential components of forest inventory analysis, following well-established allometric equations and IPCC guidelines.
- **Scaling and Data Integration:** Uniform per hectare basis enables comparability across different plot designs and supports national-level forest stock assessments.

This structured workflow and the underpinning formulas combine field inventory methodologies and data processing approaches from the NFI report to produce a robust training dataset suitable for ecological analysis, carbon accounting, or machine learning applications.